and in rapid-fire English, Misner described the theorem that Penrose had proved a few months earlier. If Penrose's theorem was right, then Khalatnikov and Lifshitz must be wrong.

The Soviet delegation was confused and incensed. Misner's English was too fast to follow, and since Penrose's theorem relied on topological arguments that were alien to relativity experts, the Soviets regarded it as suspect. By contrast, the Khalatnikov–Lifshitz analysis was based on tried-and-true methods. Penrose, they asserted, was probably wrong.

During the next few years, relativity experts in East and West plumbed the depths of Penrose's analysis, and of the Khalatnikov–Lifshitz analysis. At first both analyses looked suspect; both had dangerous, potential flaws. Gradually, however, as the experts began to master and extend Penrose's topological techniques, they became convinced that Penrose was right.

In September 1969, while I was a visiting member of Zel'dovich's research team in Moscow, Evgeny Lifshitz came to me with a manuscript that he and Khalatnikov had just written. "Please, Kip, take this manuscript back to America for me and submit it to *Physical Review Letters*", he requested. He explained that any manuscript written in the U.S.S.R., regardless of its content, was automatically classified secret until declassified, and declassification would take three months. The ludicrous Soviet system permitted me or any other foreign visitor to read the manuscript while in Moscow, but the manuscript should not itself leave the country until passed by the censors. This manuscript was too precious, too urgent for such a ridiculous delay. It contained, Lifshitz explained to me, their capitulation, their confession of error: Penrose was right; they were wrong. In 1961 they had been unable to find, among the solutions to Einstein's field equation, any singularity with completely random deformations; but now, spurred by Penrose's theorem, they and a graduate student, Vladimir Belinsky, had managed to find one. This new singularity, they thought, must be the one that terminates the implosion of randomly deformed stars and that might someday destroy our Universe at the end of the big crunch. [And, indeed, in 1993 I think they probably were right. To this 1993 viewpoint, and to the nature of their new *BKL* ("Belinsky–Khalatnikov–Lifshitz") singularity, I shall return near the end of this chapter.]